

validated ground-truth data.

To achieve the results, recordings were broken into individual images and then software tools were applied to present and edit the images very rapidly and accurately. Some technologies and techniques used for this solution were:

1. Distributed software applications
2. Distributed databases
3. HADOOP clusters
4. State-of-the-art JavaScript frameworks for the user interface, using three different backend approaches: based on Windows and using C#, based on Linux and Java, and based on Linux and Python

The complex distributed solution comprised three main components: a server backend, database and Web client application, all designed to allow for interaction with human testers.

The server backend is responsible for connecting and getting input data from company servers. The server application streams data from a company server, separates streamed data into smaller pieces using the company's API, assigns frames to individual clients, serves more than 100 client applications with requested data, aggregates output results from Web clients and sends the results to company servers. Client applications can be configured to send data in pre-defined sequences such as time-continuous or random.

The database is used by the server backend to store multiple types of data, including Big Data from numerous environmental sensors.

The Web client application enables the human tester to validate the target features identified in the current frame.

Putting together this solution requires engineers to use and coordinate multiple technologies including:

1. C#/.Net 4.5
2. Entity Framework
3. Log4Net
4. XML
5. MD5 encryption for DB credential
6. Windows Server 2012 R2

7. HTML5
8. CSS
9. JavaScript
10. Model/View/View Model→JSON/React (Bootstrap)/Backbone
11. NodeJS

Although this particular solution is designed to meet the specific needs of advanced driver assistance systems and ground-truth validation, similar technologies, methodologies and technical capabilities can be used to develop other complex sensor based and machine learning applications. The most common applications could include solutions for the IoT.

Car manufacturers have made significant advances in the past decade towards making self-driving cars a reality. However, there still remain a number of technological barriers that manufacturers must overcome before self-driving vehicles are safe enough for road use.

GPS can be unreliable, computer vision systems have limitations to understanding road scenes and variable weather conditions can adversely affect the ability of onboard processors to adequately identify or track moving objects. Self-driving vehicles are also yet to demonstrate the same capabilities as human drivers in understanding and navigating unstructured environments such as construction zones and accident areas.

These barriers, though, are not insurmountable. The amount of road and traffic data available to these vehicles is increasing, newer range sensors are capturing more data and algorithms for interpreting road scenes are evolving. Transition from human-operated vehicles to fully self-driving cars will be gradual, with vehicles at first performing only a sub-set of driving tasks such as parking and driving in stop-and-go traffic autonomously. As the technology improves, more driving tasks can be reliably outsourced to the vehicle. **EFY**



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